

The distance between two points; the equation of a circle

Pythagoras, written in coordinates — and the equation that a circle turns out to be.

LESSON 35–36

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 32–33

STAGE 9 · 3.1, 7.1

LESSON CLOCK

6'

12'

12'

6'

4'

Recall Pythagoras	6 min
The distance formula	12 min
The equation of a circle	12 min
Inside, on or outside	6 min
Homework	4 min

LEARNING OBJECTIVES

- Find the distance between two points from their coordinates.
- Recognise $(x - a)^2 + (y - b)^2 = r^2$ as the equation of a circle.
- Write the equation of a circle from its centre and radius, and read them back.
- Decide whether a given point lies inside, on or outside a circle.

TEXTBOOK

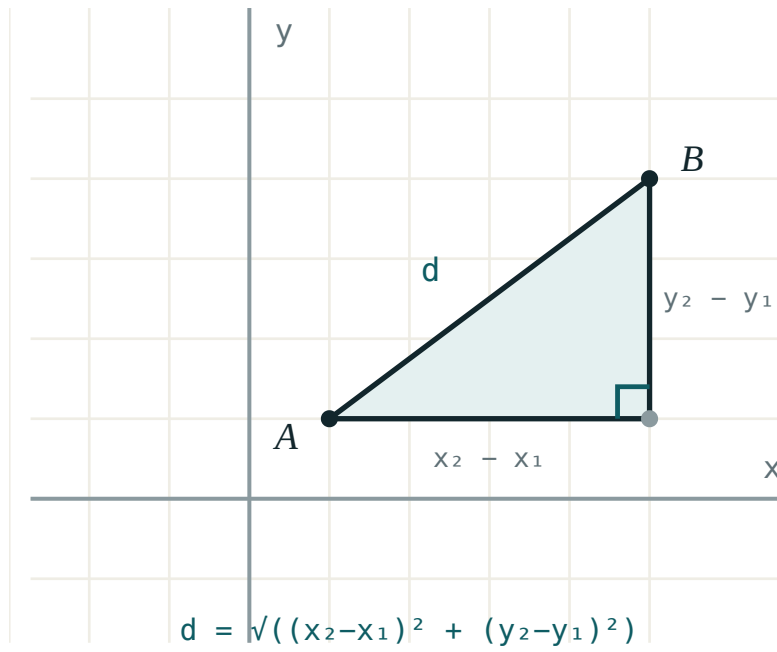
National	Темы 32–33, pp. 72–74
Cambridge	Learner’s Book pp. 56–60, 142–148
Workbook	Workbook 3.1

01

Explanation

Distance is Pythagoras in disguise

Join $A(x_1; y_1)$ to $B(x_2; y_2)$ and complete a right-angled triangle with legs parallel to the axes. The horizontal leg is $x_2 - x_1$, the vertical leg is $y_2 - y_1$, and the segment AB is the hypotenuse.



The distance between two points is the hypotenuse of a right triangle whose legs are the differences of the coordinates.

DISTANCE FORMULA

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The squares make the order of subtraction irrelevant: $(3 - 7)^2 = (7 - 3)^2 = 16$. That is the one place where this formula forgives you — everywhere else, be careful.

Leave the answer in surd form unless a decimal is asked for. $\sqrt{20} = 2\sqrt{5}$ is exact; 4.47 is not.

What a circle is, as an equation

A circle of radius r centred at $C(a; b)$ is the set of all points at distance exactly r from C . Write that with the distance formula:

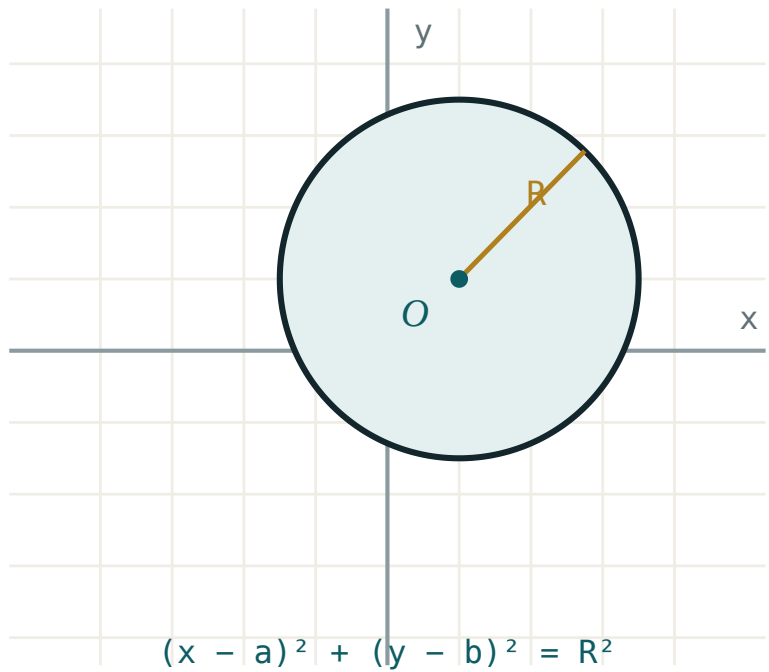
$$\sqrt{(x - a)^2 + (y - b)^2} = r$$

Square both sides — both are positive, so nothing is lost:

EQUATION OF A CIRCLE

$$(x - a)^2 + (y - b)^2 = r^2$$

Centred at the origin this becomes simply $x^2 + y^2 = r^2$.



Every point of the circle is at the same distance r from the centre — that single condition is the equation.

READ THE SIGNS BACKWARDS

$(x - 3)^2 + (y + 2)^2 = 25$ has centre $(3; -2)$, not $(3; 2)$, and radius 5, not 25. The formula subtracts the centre, so the sign you see is the opposite of the sign you want.

Inside, on, or outside

Compare the distance from the centre with the radius:

CONDITION	POSITION OF THE POINT
$(x - a)^2 + (y - b)^2 < r^2$	inside the circle
$(x - a)^2 + (y - b)^2 = r^2$	on the circle
$(x - a)^2 + (y - b)^2 > r^2$	outside the circle

There is no need to take a square root — just substitute the point into the left-hand side and compare with r^2 . Is $(6; 1)$ inside $x^2 + y^2 = 36$? Substituting gives $36 + 1 = 37 > 36$, so it is just outside.

02 Worked examples

EXAMPLE 1 Find the distance between $A(1; 2)$ and $B(5; 5)$.

① $x_2 - x_1 = 4$
The horizontal leg.

② $y_2 - y_1 = 3$
The vertical leg.

③ $AB = \sqrt{16 + 9} = \sqrt{25}$

④ $AB = 5$
A 3–4–5 triangle.

ANSWER

5

EXAMPLE 2 Write the equation of the circle with centre $(-2; 3)$ and radius 4.

① $a = -2, b = 3, r = 4$

② $(x - (-2))^2 + (y - 3)^2 = 4^2$
Substitute into the standard form.

③ $(x + 2)^2 + (y - 3)^2 = 16$
Two minus signs make a plus.

ANSWER

$$(x + 2)^2 + (y - 3)^2 = 16$$

EXAMPLE
3**A circle has equation $(x - 1)^2 + (y + 4)^2 = 25$. State its centre and radius, and say whether $(4; 0)$ lies on it.****1** centre $(1; -4)$

Change the sign of what is subtracted.

2 $r = 5$ $25 = 5^2$.**3** $(4 - 1)^2 + (0 + 4)^2 = 9 + 16 = 25$

Substitute the point.

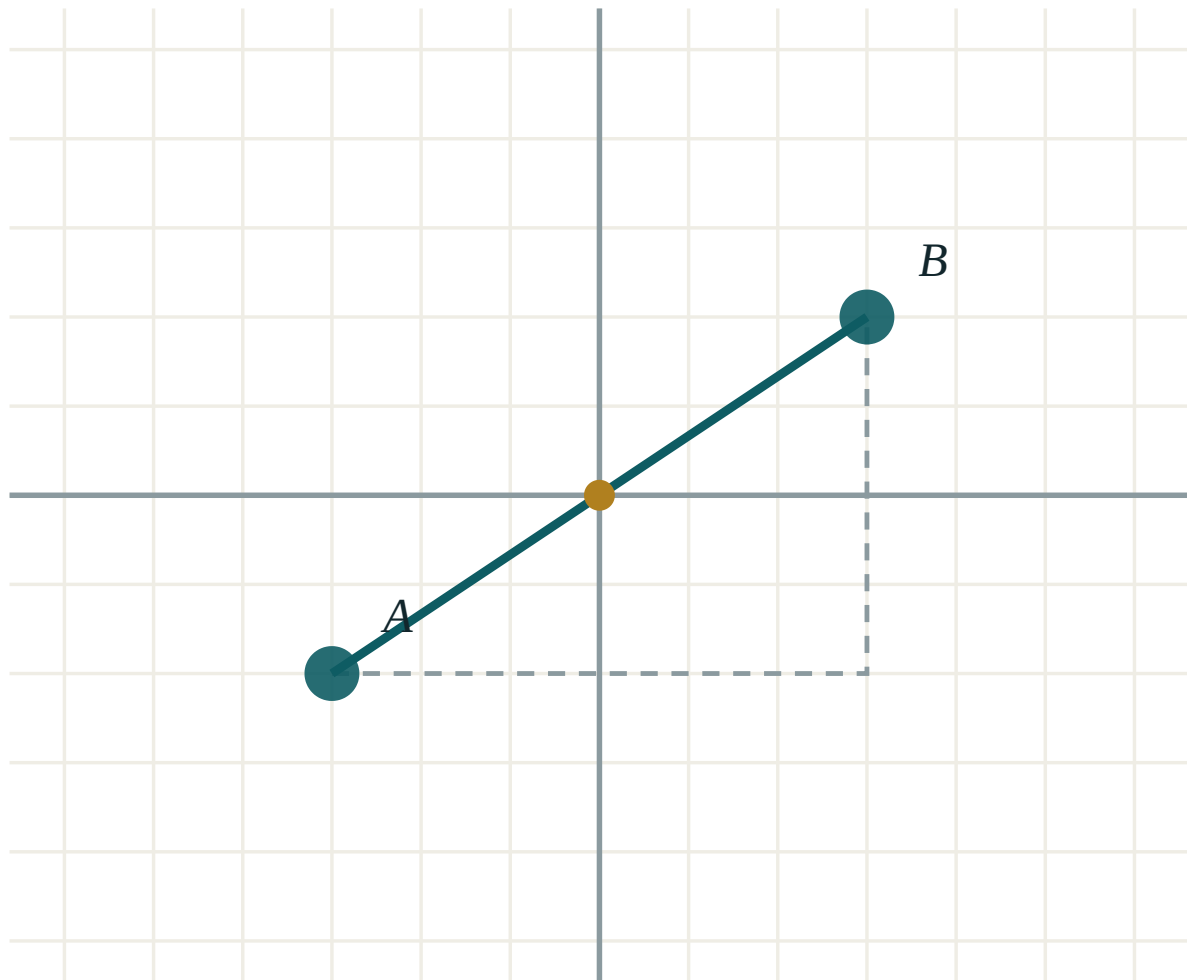
4 $25 = r^2$

So the point lies exactly on the circle.

ANSWERCentre $(1; -4)$, radius 5; the point is on the circle**03 Interactive model**

Drag A and B and watch the distance readout follow the two coordinate differences.

Distance between two points



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Distance	Masofa	Расстояние
Distance formula	Masofa formulasi	Формула расстояния
Right-angled triangle	To'g'ri burchakli uchburchak	Прямоугольный треугольник
Circle	Aylana	Окружность
Centre	Markaz	Центр
Radius	Radius	Радиус
Equation of a circle	Aylana tenglamasi	Уравнение окружности
Locus	Geometrik o'rin	Геометрическое место точек
Inside / outside	Ichida / tashqarisida	Внутри / снаружи
Unit circle	Birlik aylana	Единичная окружность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The distance between $(0; 0)$ and $(6; 8)$ is:

The centre of $(x - 2)^2 + (y + 5)^2 = 9$ is:

The radius of $x^2 + y^2 = 49$ is:

The distance between $(1; 1)$ and $(4; 5)$ is:

5

7

$\sqrt{7}$

25

Is $(3; 3)$ inside $x^2 + y^2 = 25$?

Yes

No, it is on it

No, it is outside

Cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the distance between $(0; 0)$ and $(3; 4)$.

ANSWER 5

2. Find the distance between $(1; 0)$ and $(1; 7)$.

ANSWER 7

3. Find the distance between $(2; 3)$ and $(6; 6)$.

ANSWER 5

4. Write the equation of the circle, centre $(0; 0)$, radius 6.

ANSWER $x^2 + y^2 = 36$

5. State the centre and radius of $(x - 4)^2 + (y - 1)^2 = 16$.

ANSWER $(4; 1), r = 4$

6. State the radius of $x^2 + y^2 = 100$.

ANSWER 10

7. Is $(0; 0)$ inside $x^2 + y^2 = 9$?

ANSWER Yes — $0 < 9$

Medium · the standard question

7 PROBLEMS

1. Find the distance between $(-3; 2)$ and $(1; -1)$.

ANSWER 5

2. Find the distance between $(-2; -5)$ and $(4; 3)$.

ANSWER 10

3. Write the equation of the circle, centre $(3; -2)$, radius 7.

ANSWER $(x - 3)^2 + (y + 2)^2 = 49$

4. State the centre and radius of $(x + 5)^2 + (y - 3)^2 = 36$.

ANSWER $(-5; 3)$, $r = 6$

5. Find the exact distance between $(0; 0)$ and $(2; 3)$.

ANSWER $\sqrt{13}$

6. Does $(5; 12)$ lie on $x^2 + y^2 = 169$?

ANSWER Yes — $25 + 144 = 169$

7. The centre of a circle is $(2; 2)$ and it passes through $(2; 7)$. Find its equation.

ANSWER $(x - 2)^2 + (y - 2)^2 = 25$

Hard · stretch and reason

7 PROBLEMS

1. Show that $(1; 2)$, $(4; 6)$, $(8; 3)$ form an isosceles triangle.

ANSWER $AB = 5$, $BC = 5$, $AC = \sqrt{50}$ — two sides equal

2. Show that $(0; 0)$, $(4; 0)$, $(4; 3)$ is right-angled.

ANSWER $16 + 9 = 25$ — Pythagoras holds, so the angle at $(4; 0)$ is 90°

3. A circle has centre $(1; -1)$ and passes through $(4; 3)$. Find its equation.

ANSWER $(x - 1)^2 + (y + 1)^2 = 25$

4. Find the equation of the circle whose diameter joins $(1; 2)$ and $(7; 10)$.

ANSWER centre $(4; 6)$, $r = 5$: $(x - 4)^2 + (y - 6)^2 = 25$

5. Where does $x^2 + y^2 = 25$ cut the axes?

ANSWER $(\pm 5; 0)$ and $(0; \pm 5)$

6. Is $(-3; 4)$ inside, on or outside $(x + 1)^2 + (y - 1)^2 = 16$?

ANSWER $4 + 9 = 13 < 16$ — inside

7. Find the perimeter of the triangle $(0; 0)$, $(6; 0)$, $(6; 8)$.

ANSWER $6 + 8 + 10 = 24$

07 Homework

Homework — 6 problems

Geometry 8, Темы 32–33, pp. 72–74. Leave surds exact unless a decimal is asked for.

- Find the distance between $(2; 1)$ and $(10; 7)$.
- Find the distance between $(-4; 3)$ and $(2; -5)$.
- Write the equation of the circle, centre $(-1; 4)$, radius 3.
- State the centre and radius of $(x - 6)^2 + (y + 2)^2 = 81$.
- Show that $(0; 0)$, $(5; 0)$, $(0; 12)$ is right-angled and find its perimeter.

6. Is $(2; 6)$ inside, on or outside $x^2 + y^2 = 40$?